

#HW 2

● Graded

Student

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 [View or edit group](#)

Total Points

100 / 100 pts

Question 1

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 2 pts There has been a mistake while you inter-converted the units. Your answer should be -4.63 Btu
- 0.5 pts Your final answer is not clear. You are supposed to write -4.63. The number 4 is not clear.
- 2 pts You were supposed to convert your answer to Btu
- 7 pts The answer has not been submitted.
- 2 pts Answer should be in Btu.

Question 2

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 7 pts Answer has not been submitted.
- 4 pts You have to find out the area of the plot. To calculate work you should multiply the volume difference with pressure. You have taken individual volumes and that should not be done.
- 2 pts You need to add up the areas. You have subtracted instead.
- 7 pts Answer has not been submitted.

Question 3

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 2 pts You have to calculate the work done in all the three steps and add them up. You have only done for the first step.
- 2 pts The work done will be negative in the first step. The total work should therefore be -2000 kJ.
- 2 pts It has been given $P(V^{1.5})$ constant. So $P = \text{constant}/(V^{1.5})$. You should apply that in integration of PdV in order to calculate work.
- 1 pt You should calculate the total work in all three steps.
- 2 pts There is an error in your calculation. The total work done will be -2000 kJ.
- 7 pts Answer has not been submitted.

Question 4

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 1 pt In part b E_2 will be 22 in order to get 15 as the difference.
- 4 pts ΔE is the difference of two energies. E_1 is to be subtracted from E_2 . You should not add E_1 and E_2 .
- 1 pt in part e, E_1 should be +8 to get ΔE 0.
- 1 pt in c, ΔE should be -14 from $Q-W$, accordingly E_2 should be -8 to get ΔE as -14.
- 7 pts Answer has not been submitted.

Question 5

(no title)

10 / 10 pts

✓ - 0 pts Correct

- 5 pts Answer has not been submitted.
- 1 pt You should calculate the total heat transfer in the process.
- 2 pts There is a mistake while you changed units. Answer should be -50kJ.

Question 6

(no title)

30 / 30 pts

✓ - 0 pts Correct

- 15 pts No essay submitted

Question assigned to the following page: [1](#)

- (a) Carbon dioxide (CO_2) gas within a piston-cylinder assembly undergoes a process from a state where $p_1 = 5 \text{ lbf/in.}^2$, $V_1 = 2.5 \text{ ft}^3$ to a state where $p_2 = 20 \text{ lbf/in.}^2$, $V_2 = 0.5 \text{ ft}^3$. The relationship between pressure and volume during the process is given by $p = 23.75 - 7.5V$, where V is in ft^3 and p is in lbf/in.^2 . Determine the work for the process, in Btu.

1)

$$P_1 = 5 \text{ lbf/in}^2$$

$$P_2 = 20 \text{ lbf/in}^2$$

$$P = 23.75 - 7.5V$$

$$V_1 = 2.5 \text{ ft}^3$$

$$V_2 = 0.5 \text{ ft}^3$$

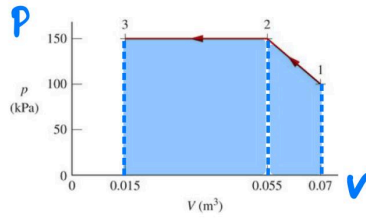
$$W = \int P dV \quad W = \int_{2.5}^{0.5} (23.75 - 7.5V) dV = 23.75V - \frac{7.5V^2}{2} \Big|_{2.5}^{0.5}$$

$$23.75(0.5) - \frac{7.5(0.5)^2}{2} - 23.75(2.5) - \frac{7.5(2.5)^2}{2}$$

$$= -25 \frac{\text{lbf} \cdot \text{ft}^3}{\text{in}^2} \left(\frac{144 \text{ in}^2}{1 \text{ ft}^2} \right) = -3600 \text{ lbf} \cdot \text{ft} \left(\frac{1 \text{ BTU}}{778.17 \text{ lbf} \cdot \text{ft}} \right) = \boxed{-4.63 \text{ BTU}}$$

Question assigned to the following page: [2](#)

- (a) Air contained within a piston-cylinder assembly is slowly compressed. As shown in the figure below, during this process the pressure first varies linearly with volume and then remains constant. Determine the total work, in kJ.



$$2) W = \int_{0.015}^{0.07} P dv = \int_{0.015}^{0.055} P dv + \int_{0.055}^{0.07} P dv$$

$$= (0.055 - 0.015 \text{ m}^3)(150 \text{ kPa}) + (0.07 - 0.055 \text{ m}^3) \left(\frac{150 \text{ kPa} + 100 \text{ kPa}}{2} \right) \\ - 1 (6.00 \text{ kPa} \cdot \text{m}^3 + 1.875 \text{ kPa} \cdot \text{m}^3) = -7.88 \text{ kPa} \cdot \text{m}^3$$

$$-7.88 \text{ kPa} \cdot \text{m}^3 \left(\frac{10^3 \text{ Pa} \cdot \text{m}^3}{1 \text{ kPa} \cdot \text{m}^3} \right) \times \left(\frac{1 \text{ J}}{1 \text{ Pa} \cdot \text{m}^3} \right) \times \left(\frac{10^{-3} \text{ kJ}}{1 \text{ J}} \right) = \boxed{-7.88 \text{ kJ}}$$

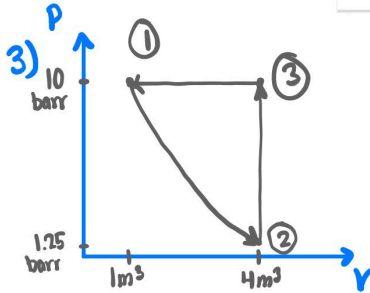
Question assigned to the following page: [3](#)

EXERCISE 2.13.24: (Problem 2.24 in the 9th edition).

(a) Carbon dioxide (CO_2) gas in a piston-cylinder assembly undergoes three processes in series that begin and end at the same state (a cycle).

- **Process 1-2:** Expansion from state 1 where $p_1 = 10 \text{ bar}$, $V_1 = 1 \text{ m}^3$, to state 2 where $V_2 = 4 \text{ m}^3$. During the process, pressure and volume are related by $pV^{1.5} = \text{constant}$.
- **Process 2-3:** Constant volume heating to state 3 where $p_3 = 10 \text{ bar}$.
- **Process 3-1:** Constant pressure compression to state 1.

Sketch the processes on p - V coordinates and evaluate the work for each process, in kJ. What is the net work for the cycle, in kJ?



$$p_1 = 10 \text{ barr} \quad V_1 = 1 \text{ m}^3 \quad \text{A}$$

$$p_2 = 1.25 \text{ barr} \quad V_2 = 4 \text{ m}^3 \quad \text{B}$$

$$pV^{1.5} = C$$

$$p_3 = 10 \text{ barr} \quad V_3 = 4 \text{ m}^3 \quad \text{C}$$

$$p_4 = 10 \text{ barr} \quad V_1 = 1 \text{ m}^3$$

$$pV^{1.5} = 10 \cdot 1^{1.5}$$

$$A) \quad W = \int_1^4 \frac{10 \cdot 1^{1.5}}{V^{1.5}} dV = 10 \text{ barr} \cdot \text{m}^3 \left(\frac{100 \text{ kPa}}{1 \text{ barr}} \right) = 1000 \text{ kJ}$$

$$B) \quad W = \int_{V_1}^{V_2} p dV \quad V_1 = V_2 \quad \int = 0 \text{ kJ}$$

$$C) \quad W = \int_4^1 10 dV = 10V \Big|_4^1 = 10(1 \text{ m}^3) - 10(4 \text{ m}^3) = -30 \text{ barr} \cdot \text{m}^3 = -3000 \text{ kJ}$$

$$\text{Work TOTAL} = \boxed{-2000 \text{ kJ}}$$

Question assigned to the following page: [4](#)

**EXERCISE**

2.13.43: (Problem 2.43 in the 9th edition). ?

- (a) Each line of the following table gives data, in Btu, for a process of a closed system. Determine the missing table entries, in Btu.

Process	Q	W	E_1	E_2	ΔE
a	-10	-25	+15	+30	+15
b	+27	12	+7	22	+15
c	-4	+10	+6	-8	-14
d	10	-10	-10	+10	+20
e	+3	3	8	+8	0

 $(30-15)$

4) $\Delta E = E_f - E_i = Q - W$

a) $30 - 15 = \Delta E$ $15 = Q - (-25)$ $Q = -10$

b) $15 = 27 - W$ $W = 12$ $15 = E_2 - E_1$ $E_2 = 22$

c) $-4 - 10 = -14 = \Delta E$ $-14 = E_2 - 6$ $E_2 = -8$

d) $20 = 10 - E_1$ $E_1 = -10$ $20 = Q - (-10)$ $Q = 10$

e) $0 = 8 - E_1$ $E_1 = 8$ $0 = 3 - W$ $W = 3$

1

Question assigned to the following page: [5](#)

5)



EXERCISE

2.13.44: (Problem 2.44 in the 9th edition). ?

- (a) A closed system of mass 10 kg undergoes a process during which there is energy transfer by work from the system of 0.147 kJ per kg, an elevation decrease of 50 m, and an increase in velocity from 15 m/s to 30 m/s. The specific internal energy decreases by 5 kJ/kg and the acceleration of gravity is constant at 9.7 m/s². Determine the heat transfer for the process, in kJ.

$$g = 9.7 \text{ m/s}^2$$

$$u = 5 \text{ kJ/kg} \downarrow$$

$$m = 10 \text{ kg}$$

$$w = 0.147 \text{ kJ/kg}$$

$$W = 1.47 \text{ kJ}$$

$$50 \text{ m} \downarrow 15 \text{ m/s} \rightarrow 30 \text{ m/s}$$

$$A) \Delta E = Q - W$$

$$\Delta KE + \Delta PE + \Delta U = Q - W$$

$$\Delta \frac{1}{2}mv^2 + \Delta mgh + \Delta U = Q - W$$

$$\left[\frac{1}{2}(10 \text{ kg})(30 \frac{\text{m}}{\text{s}})^2 - \frac{1}{2}(10 \text{ kg})(15 \frac{\text{m}}{\text{s}})^2 \right] + [(10 \text{ kg})(9.7 \frac{\text{m}}{\text{s}^2})(-50)] + [-50 \text{ kJ}]$$

$$3375 \text{ J} + -4850 \text{ J} + -50 \text{ kJ} = Q - W$$

$$-51.475 \text{ kJ} + 1.47 \text{ kJ} = Q$$

$$Q = -50.005$$

Question assigned to the following page: [6](#)

One of my current role models in chemical engineering is Regina M. Murphy. This semester I've finally been immersed in the chemical engineering department at UIC as it's my first term actually taking department courses. So far I like the content I'm learning, and I've definitely felt the jump in difficulty from gen ed classes to engineering courses. I chose Regina M. Murphy as my role model because she's the author of the book we use in Chemical Engineering 210. As I've read the book for class, I've found it interesting and well thought out. Even in the preface where she talks about her mission she really shows how carefully she approached writing it.

Regina M. Murphy earned her B.S. in chemical engineering at MIT. As an undergrad, she played varsity softball. After graduation, she worked at the Chevron Richmond Refinery to gain industry experience. Five years later, she returned to MIT to complete her Ph.D. She became the first female faculty hire in chemical engineering at the University of Wisconsin–Madison, and later the first female department chair in the university's history. She has worked as a professor for more than three decades and recently retired. She also served on many campus committees, including the Physical Sciences Divisional Committee.

She redeveloped the chemical engineering curriculum at her university. Looking at the course outline and schedule, it flows well, almost similar to UIC in some ways and different in others. For example, they offer an intro to chemical engineering course that students can take as freshmen to see if the field is right for them. While UIC doesn't offer that exact course. (I know the biomedical engineering department offers something like it, as that was one reason for my major switch.) Besides curriculum work, she wrote a textbook with McGraw Hill in 2007 that became a foundation for introduction to chemical processes. If it weren't a good book, we wouldn't still be using it today, as it's only a year younger than me, 18 years old and still relevant. You can tell she put real effort into making the book helpful, engaging, and interactive. She has also won many awards, for example the R. Byron Bird Chair.

Question assigned to the following page: [6](#)

I chose her because I don't know many people who've had a big impact in chemical engineering. She wrote the book we're using in Chemical Engineering 210, and I think it's well written. It has taught me a lot, even though the course structure is definitely rigorous compared with past coursework. I connect with her story on how she was the first at many things. The first female faculty member in her department, and the first female department chair. She broke barriers and set new standards. I'm the first in my family to go to college, and when I graduate I'll be the first with my last name to hold a bachelor's degree. She broke glass ceilings in her department, and I want to do the same in my family.

She's had a major impact on the chemical engineering community through the books she's written. We're using one of them from about 18 years ago, which I would conclude that many classes before mine used it, too. When you talk to chemical engineers, they probably know the book or at least its approach.

All in all, I chose her because she's a trailblazer in her field and in her department by redefining her program, influencing others, and writing books that help teach past, current, and future generations of chemical engineering students.

Question assigned to the following page: [6](#)

References (APA)

- ELY, S. E., & ELY, S. E. (2024, June 14). *Regina Murphy retires from CBE - College of Engineering - University of Wisconsin-Madison*. College of Engineering - University of Wisconsin-Madison. <https://engineering.wisc.edu/blog/regina-murphy-retires-from-cbe/>
- Murphy, Regina - UW-Engineering Directory | College of Engineering @ The University of Wisconsin-Madison*. (2025). Wisc.edu. https://directory.engr.wisc.edu/cbe/Faculty/Murphy_Regina/
- Wikipedia Contributors. (2024, August 22). *Regina Murphy*. Wikipedia; Wikimedia Foundation. https://en.wikipedia.org/wiki/Regina_Murphy

a. Introduction (100 words) - Provide a brief overview of your chosen role model and why you find them inspiring.

b. Biographical Information (200 words) - Present key biographical details about your role model, including their name, birthplace, education, and career path.

c. Contributions to Chemical Engineering (300 words) - Describe in detail the major contributions, innovations, or breakthroughs that your role model has made in the field of chemical engineering. Explain the significance of these contributions to the discipline.

d. Personal Connection (200 words) - Share your personal reasons for choosing this individual as your role model. Discuss the aspects of their life, work, or values that resonate with you and inspire you.

e. Impact on the Discipline (200 words) - Analyze how your role model's work has influenced the field of chemical engineering. Discuss any lasting legacies, advancements, or changes they have brought about.

f. Conclusion (100 words) - Sum up your report by reiterating why your chosen role model is significant in the field of chemical engineering and how they serve as an inspiration to you. Do not forget to Cite Your Sources: Ensure that you properly cite all the sources you have used for your research following a recognized citation style (e.g., APA, MLA, or Chicago).